Week 1 — Course Introduction

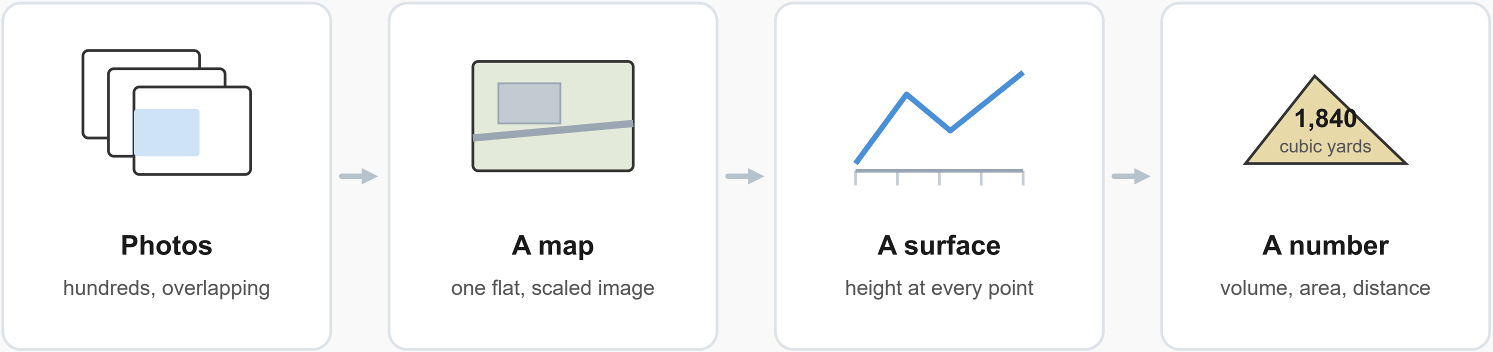
CCE 194R — Aerial Measurement

A drone is a measuring instrument

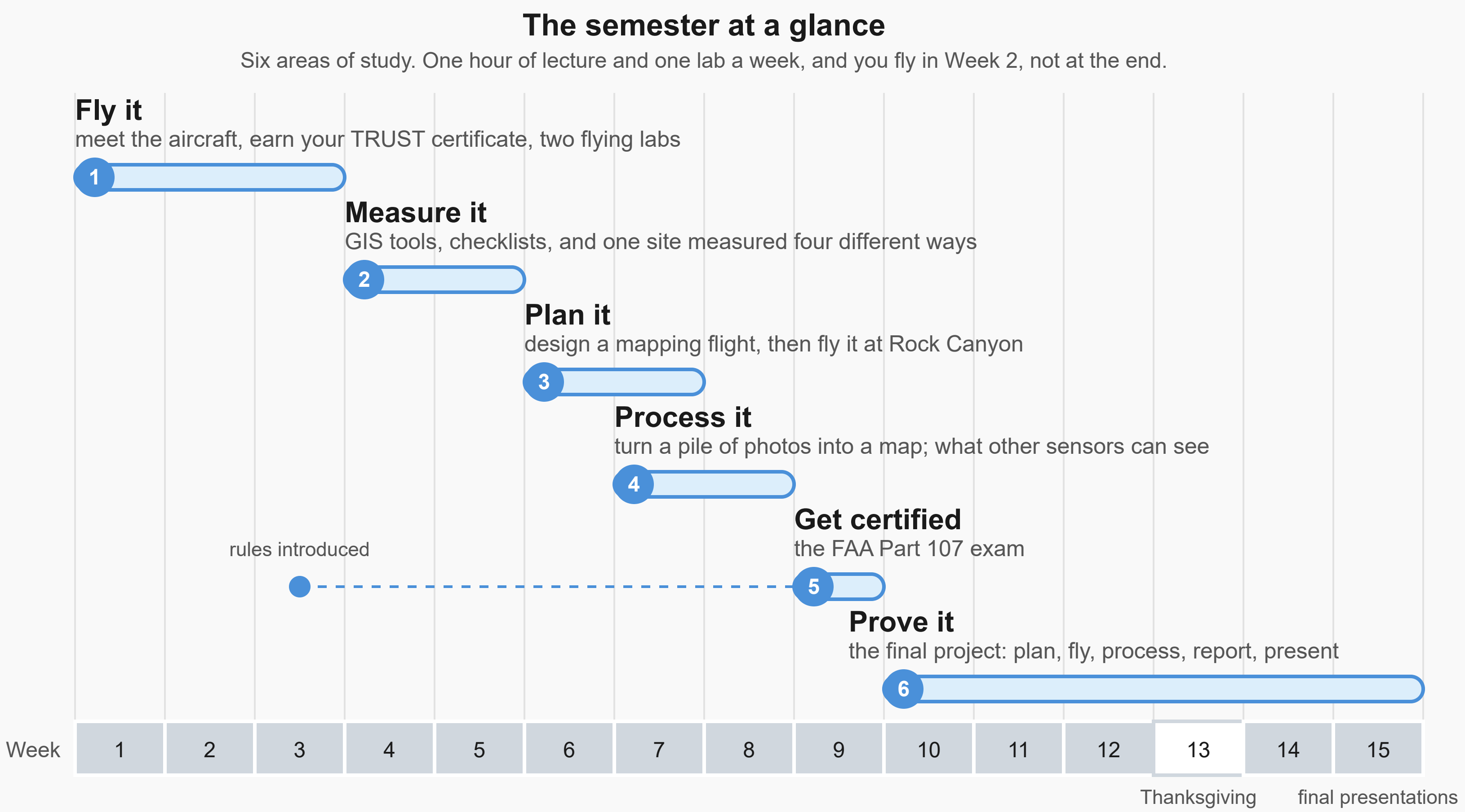
It happens to fly. What matters is the data it brings back.

One flight, four things you can use

The same set of photos becomes each of these in turn.



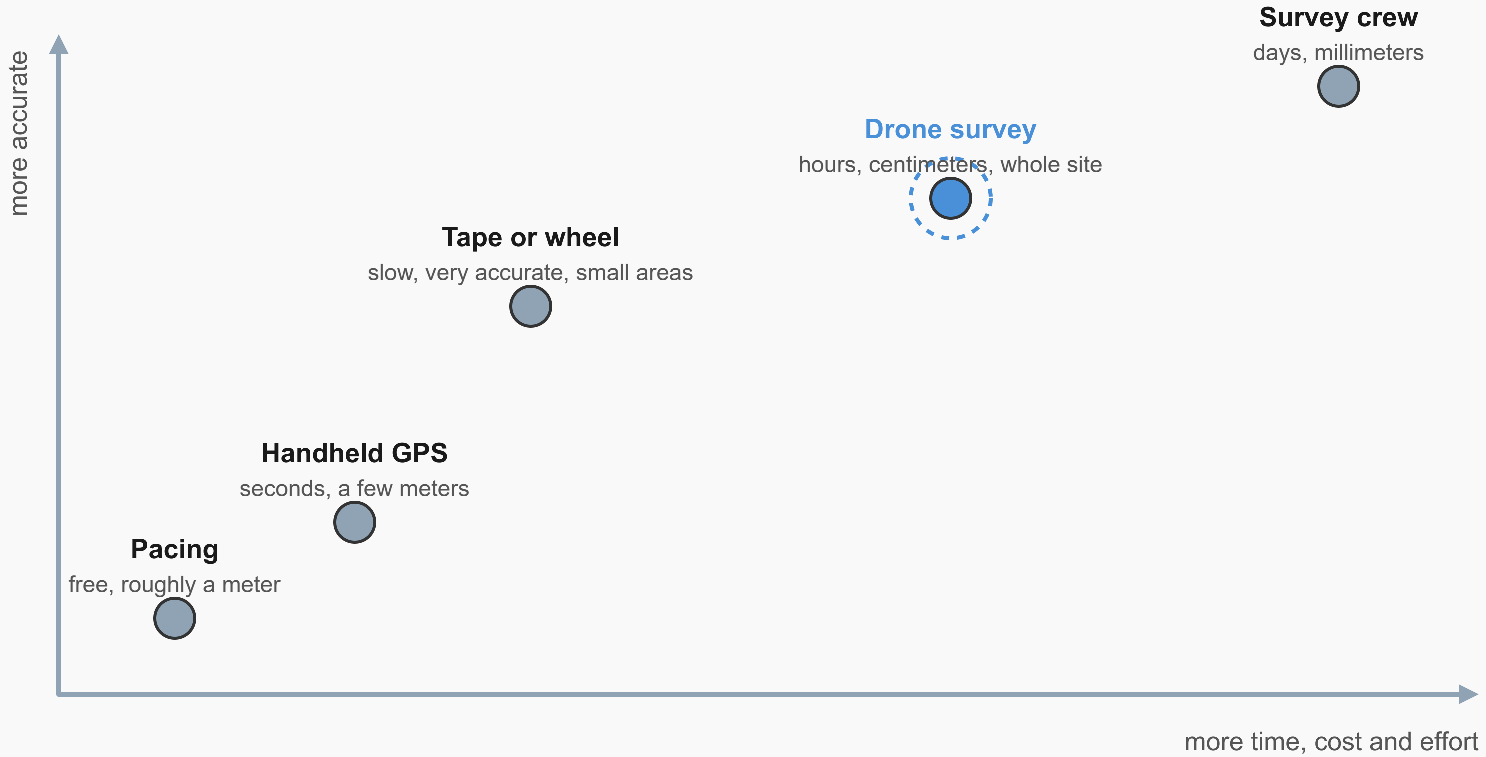
One flight, several products — each answers a different question.



Six areas of study across the semester. You fly in Week 2, not at the end.

Picking a measurement method

There is no best method, only the cheapest one that still answers the question.



This course is about knowing which dot to reach for, and being able to explain why.

There is no best method, only the cheapest one that still answers the question.

The right tool for the job

Groups of 3. For each scenario, pick a method and say why. 8 minutes.

Eyeball/pace < tape < Google Maps/Earth < drone photos < drone + ground control < survey crew

Scenarios 1–5

1. Site recon — does a 100-acre field have boulders or trees on it?
2. Structural monitoring — has a bridge support settled more than 5 mm this year?
3. Earthwork volume — bill a 50-ft stockpile that changes size weekly.
4. New flooring — how much laminate for a bedroom at home?
5. Property dispute — neighbor says your fence is 2 ft onto their land.

Scenarios 6–9

6. Parking count — is this gravel lot big enough for 40 cars?
7. Progress docs — show a 20-acre site every two weeks, for the owner.
8. Floor flatness — slab must meet 1/8 inch over 10 feet.
9. Roof damage — after a hailstorm, what fraction of the roof is damaged?

Answers 1–3

1. **Google Earth** (or a fresh drone flight) — a yes/no question, not a number.
2. **Survey crew**, precise levelling — below drone-map resolution; the answer keeps a bridge open.
3. **Drone map with ground control** — a few percent is fine for billing; a tape can't measure a pile.

Answers 4–6

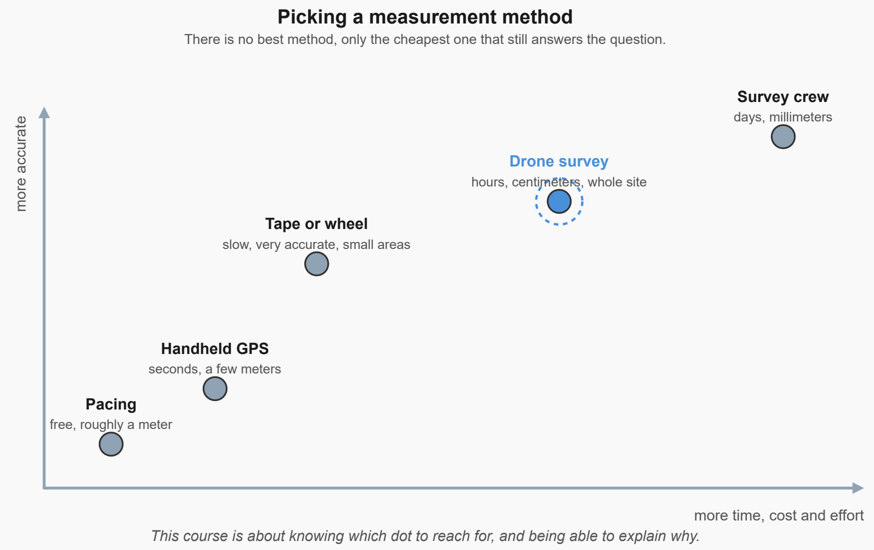
4. **Tape measure** — an inch either way changes nothing; you order extra anyway.
5. **Licensed surveyor**, total station or survey-GPS — a legal boundary must be defensible in court.
6. **Pacing, or Google Maps** — a rough area is fine; two minutes, not two hours.

Answers 7–9

7. **Drone photos**, no ground control — the owner wants to see progress, not measure it.
8. **Straightedge and feeler gauges**, or a laser level — a millimetre question over a short distance.
9. **Drone photos** — a few percent is fine, and it keeps a person off a damaged roof.

The pattern

Most accuracy went to the most expensive tool, least to the cheapest — the drone won the middle.



Accuracy rises with effort, but not smoothly — pick the method the decision needs.

Before you fly: TRUST

Required before Week 2's flying lab. Free, about 30 minutes, you cannot fail it.

Download your certificate and submit it in Learning Suite.

How the site works

The left nav is organized by week. Each week has a "This Week" page like this one.

Due dates are not on this site — see Learning Suite.

Before next week

- Read Welcome to Drone Measurements
- Take the syllabus quiz in Learning Suite
- Earn your TRUST certificate
- Week 2 is the first flying lab